

Author Guidelines for the British Machine Vision Conference

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Abstract

Automated baggage screening in security-critical environments faces persistent challenges including extreme class imbalance, limited availability of labeled anomaly samples, and the inherent difficulty of detecting small, heavily occluded threat objects within cluttered X-ray imagery. While supervised deep learning approaches achieve strong detection accuracy, they depend on extensive annotated datasets and frequently fail to generalize to novel or unseen threat types. We propose a **Two-Stage Reconstruction-Guided Hybrid Framework** that combines unsupervised representation learning with lightweight supervised classification for both anomaly detection and spatial localization. In the first stage, a Convolutional Autoencoder (CAE) is trained exclusively on normal baggage images; the resulting pixel-wise reconstruction error, the disparity map serves as the primary anomaly signal, from which discriminative features are extracted to train a Random Forest classifier for image-level detection. In the second stage, a patch-wise reconstruction strategy enables fine-grained spatial analysis, with connected component analysis applied to the thresholded disparity mask to identify and refine anomalous regions as padded bounding boxes. Evaluated on the large-scale SIXray benchmark, the proposed framework achieves a Detection ROC-AUC of 0.916, a detection accuracy of 99.4%, and a localization accuracy of 0.908 at $\text{IoU} \geq 0.30$, outperforming GANomaly, Skip-GANomaly, and Dense-GANomaly baselines while remaining computationally efficient for real-world deployment.

1 Introduction

The security of modern transportation infrastructure depends on the reliable detection of prohibited items within passenger baggage. X-ray screening remains the primary non-intrusive mechanism for this task, enabling inspection of opaque containers without physical search [8]. However, the scale of modern transit operations has made purely manual interpretation untenable: security operators must analyze thousands of cluttered, noisy images per shift under sustained cognitive load, and their attention degrades accordingly [9]. Transitioning to Automated Threat Recognition (ATR) systems has therefore become an operational imperative.

The dominant paradigm in the literature applies supervised deep learning to this problem, adapting established object detection architectures such as YOLO and Faster R-CNN to the X-ray domain [1]. These approaches have demonstrated high detection rates for specific known threat categories. However, they are fundamentally constrained by two interlocking problems. First, the class imbalance in operational environments is extreme: prohibited items appear in a vanishingly small fraction of scans [1], causing supervised models to bias heavily towards the majority (benign) class and produce unacceptably high false-negative rates on rare threats. Second, supervised models operate under a *closed-set* assumption, meaning they can only detect threat types present in their training data. Novel improvised devices, non-metallic 3D-printed firearms, liquid explosives, or concealed components fall outside predefined categories and are effectively invisible to such systems [1]. Acquiring large, balanced, and annotated datasets of real prohibited items is additionally expensive and logistically dangerous.

Unsupervised Anomaly Detection (UAD) offers a compelling alternative by reframing the problem: rather than recognizing specific threats, the system learns what *normal* baggage looks like and flags statistical deviations for human review. Reconstruction-based methods, notably GANomaly [2] and Skip-GANomaly [3], exploit the intuition that an autoencoder trained exclusively on normal images will fail to faithfully reconstruct anomalous content, producing an elevated reconstruction error at the location of the threat. While this approach is theoretically well-motivated, existing adversarial reconstruction methods carry significant computational overhead and, critically, none provide a principled mechanism for precise spatial localization of detected anomalies within the image, a capability that is essential for guiding operator attention in practice.

In this paper, we address these limitations with a **Two-Stage Reconstruction-Guided Hybrid Framework** designed for both accurate image-level detection and fine-grained anomaly localization on X-ray security imagery. Our contributions are as follows:

1. **Reconstruction-Driven Feature Extraction for Classification.** We train a Convolutional Autoencoder solely on normal baggage images and derive a pixel-wise disparity map as the primary anomaly signal. Statistical features extracted from this map are used to train a Random Forest classifier, combining the generalization of unsupervised representation learning with the discriminative power of a supervised classifier, without requiring annotated anomaly samples at training time.
2. **Patch-Wise Localization via Connected Component Analysis.** When an anomaly is detected at the image level, a patch-wise reconstruction strategy applies the autoencoder at fine spatial granularity. A threshold is applied to the resulting disparity mask to produce a binary anomaly map, from which connected components are extracted, size-filtered, and padded to form bounding box predictions.
3. **Comprehensive Evaluation on SIXray.** We benchmark against GANomaly, Skip-GANomaly, and Dense-GANomaly on the SIXray dataset [1], demonstrating that our framework achieves superior detection accuracy (99.4%) and ROC-AUC (0.916), while additionally providing localization performance ($\text{IoU} \geq 0.30$ accuracy of 0.908) that prior baselines do not report.

The remainder of this paper is organized as follows. Section ?? reviews related work in classical, supervised, and unsupervised X-ray threat detection. Section ?? details the proposed two-stage architecture and its loss formulation. Section ?? describes the experimental

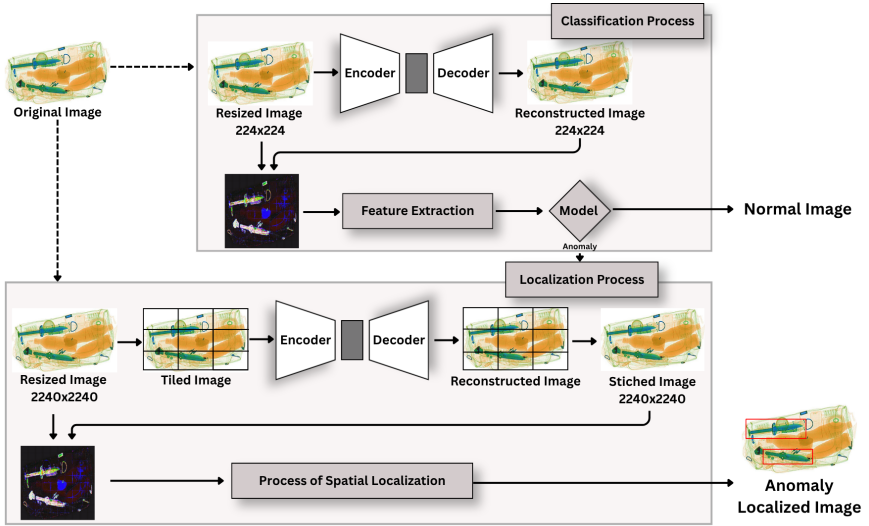


Figure 1: Two stage pipeline

setup, datasets, and evaluation metrics. Section ?? presents quantitative and qualitative results, and Section 4 concludes with directions for future work.

2 Proposed Method

2.1 Overview

The proposed framework addresses automated baggage threat detection through a two stage hybrid pipeline that combines unsupervised reconstruction based learning with supervised classification. The core insight is that a convolutional autoencoder (CAE) trained exclusively on normal X-ray images will reconstruct anomalous inputs poorly, producing elevated per pixel error in regions containing threat objects. This reconstruction error, captured as a disparity map, serves as the anomaly signal driving both image level detection and spatial localization.

The framework is motivated by two practical realities of real world screening: first, anomalous baggage content is rare and costly to annotate, making purely supervised approaches brittle; second, operators require not just a detection flag but a spatial indicator pointing to the suspect region. The first stage feeds disparity map features into a classifier to produce an image level binary decision. If an anomaly is detected, the second stage triggers a patch wise reconstruction process, followed by connected component analysis, to localize and bound the threatening region within the image.

2.2 Convolutional Autoencoder with Combined Loss

The foundation of both stages is a convolutional autoencoder trained solely on normal (threat free) X-ray images from the SIXray dataset. Because the model only learns to reconstruct the

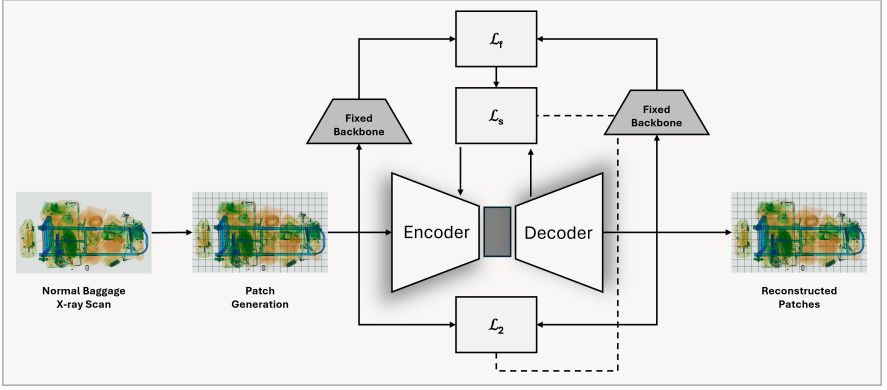


Figure 2: Autoencoder training

distribution of benign baggage content, it generalizes poorly to anomalous inputs, producing higher reconstruction error precisely where threats are present.

Architecture. The encoder consists of four convolutional blocks, each comprising a 3×3 convolution, batch normalization, and ReLU activation, followed by 2×2 max pooling to downsample the spatial resolution. Filter depth increases progressively ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256$), enabling the network to capture increasingly abstract representations of normal baggage structure at each scale. The bottleneck produces a compact latent vector encoding the essential structure of the input. The decoder mirrors the encoder symmetrically, using transposed convolutions to progressively upsample back to the original resolution of 224×224 , with a final sigmoid activation to constrain pixel values to $[0, 1]$.

Training objective. A key design choice is the use of a combined loss function that jointly optimizes pixel level fidelity and perceptual feature similarity:

$$\mathcal{L}_s = \alpha_1 \mathcal{L}_f + \alpha_2 \mathcal{L}_2 \quad (1)$$

The first term, $\mathcal{L}_f$, is a feature level loss computed in the space of a pretrained feature extractor $f(\cdot)$, measuring the mean absolute difference between the feature representations of the original and reconstructed images across a batch of size b_s :

$$\mathcal{L}_f = \frac{1}{b_s} \sum_{i=1}^{b_s} |f(k_i) - f(k'_i)| \quad (2)$$

The second term, $\mathcal{L}_2$, is a pixel wise mean squared error between the original image k_i and its reconstruction k'_i :

$$\mathcal{L}_2 = \frac{1}{b_s} \sum_{i=1}^{b_s} |k_i - k'_i|^2 \quad (3)$$

The weighting coefficients α_1 and α_2 balance the contribution of each term. This combined loss is critical: $\mathcal{L}_2$ alone tends to produce blurry reconstructions that lose high frequency details, while $\mathcal{L}_f$ enforces structural and perceptual consistency. Together, they produce reconstructions that are faithful to normal imagery while remaining sensitive to subtle anomalous structures such as small metallic objects embedded in cluttered baggage.

Disparity Map Computation: At inference time, for an input image I , the CAE produces a reconstruction R . The disparity map D is computed as the per pixel absolute difference:

$$D(x,y) = \|I - R\| \quad (4)$$

This map assigns high values to regions where the model’s reconstruction deviates from the input, the spatial signature of anomalous content. It forms the input signal to both downstream stages.

2.3 Feature Extraction and Image Level Classification

While the raw disparity map encodes spatial anomaly information, feeding it directly to a classifier would be high dimensional and noisy. Instead, a compact set of statistical features is extracted from D to produce a fixed length descriptor for downstream classification.

Feature extraction. Features are derived from the disparity map to capture both global reconstruction error magnitude and local spatial structure. These include global statistics (mean, standard deviation, maximum, and upper percentile values of D), spatial concentration metrics (the proportion of pixels exceeding an error threshold), histogram based distribution descriptors, and regional block statistics capturing localized error patterns that may correspond to small threat objects. The resulting feature vector compactly summarizes the anomaly signal in a form suitable for classical machine learning, while preserving interpretability and robustness under limited labeled data.

Classifier selection. The disparity map features are used to train a Random Forest classifier. Random Forests are well suited to this setting for three reasons: the feature vectors are low-dimensional, making deep classifiers unnecessary; the ensemble’s bootstrap aggregation provides robustness to the feature scale heterogeneity inherent in mixed statistical descriptors; and class imbalance can be directly addressed by setting tree-level class weights inversely proportional to class frequency, without requiring oversampling or data augmentation. The ensemble comprises 200 decision trees, with class weights calibrated to compensate for the severe imbalance between normal and anomalous samples in the SIXray dataset.

2.4 Patch Wise Localization via Connected Component Analysis

Image level detection indicates whether a threat is present but provides no spatial guidance. When the classification stage flags an image as anomalous, the localization stage is activated to identify the precise region of interest.

Patch wise reconstruction. Rather than analyzing the full image reconstruction holistically, the localization stage operates at fine spatial granularity. The input image is upscaled to 2240×2240 pixels and partitioned into non overlapping tiles. Each tile is independently reconstructed by the CAE, producing a tiled disparity map that retains high spatial resolution across the full image extent. This patch wise strategy allows the model to isolate reconstruction errors to small sub regions, which is critical for detecting compact threat objects that would otherwise be diluted in a global analysis.

Localization pipeline. The spatial anomaly signal is refined through the following sequence of operations:

Step 1 – Disparity Computation: The disparity map

$$D(x, y) = \|I - R\| \quad (5)$$

is computed from the tile wise reconstructions, producing a high resolution map of reconstruction error across the image.

Step 2 – Thresholding: A binary mask M is obtained by applying a threshold T :

$$M(x, y) = \begin{cases} 1, & \text{if } D(x, y) > T \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

The threshold T is optimized on the validation set by sweeping candidate values and selecting the one that maximizes localization F1 at IoU ≥ 0.3 .

Step 3 – Connected Component Extraction: Connected component analysis (CCA) is applied to the binary mask M to identify spatially contiguous regions of elevated reconstruction error. Each connected component represents a candidate threat region.

Step 4 – Component Filtering: Components are filtered by size, retaining only those whose area falls within a predefined range $[min_size, max_size]$. This removes both noise induced micro components and large background artifacts that are unlikely to correspond to localized threats.

Step 5 – Bounding Box Generation: Each surviving component is enclosed in a padded bounding box, projected back to the coordinate space of the original image, and returned as the localization output.

The hyperparameters T , min_size , and max_size are each tuned independently on the validation set. This post processing pipeline transforms the continuous disparity signal into discrete, interpretable bounding regions, directly actionable by human screening operators.

2.5 Design Rationale

Three principles guided the framework’s design:

- **Label efficiency:** The CAE requires no anomaly annotations during training, and the RF classifier requires only image level labels rather than pixel wise segmentation masks.
- **Modularity:** The classification and localization stages are decoupled and can be updated independently. For instance, the RF could be replaced with a neural classifier as labeled data accumulates, without retraining the CAE.
- **Computational practicality:** Both the feature extraction and Random Forest inference are lightweight, and the CAE architecture is kept shallow enough for deployment in time constrained security screening environments without GPU dependency at inference time.

3 Experiments

3.1 Dataset and Experimental Setup

All experiments were conducted on the SIXray dataset, a large scale benchmark for X-ray baggage inspection that contains over 1,059,231 images across six categories: gun, knife,

wrench, pliers, scissors, and a normal class. The dataset is characterised by a highly imbalanced class distribution, with anomalous images constituting a small fraction of the total, which closely mirrors real world security screening conditions.

Autoencoder training. Because the reconstruction based anomaly detection paradigm assumes the model learns an exclusive representation of normality, only normal (non threat) images were used during autoencoder training. Exposing the model to anomalous samples during training would undermine its ability to produce elevated reconstruction error on unseen threats, as the network would learn to reconstruct both normal and anomalous structures. The normal subset was split 80:20 for training and validation, respectively, with the validation split used exclusively for monitoring reconstruction loss and selecting model checkpoints.

Classification training. To construct a labelled dataset for the downstream classification stage, normal and anomalous images were sampled at a 10:1 ratio, reflecting the natural scarcity of threat items in operational settings and deliberately preserving the class imbalance challenge. This combined set was partitioned using a 70:20:10 stratified split for training, validation, and testing, respectively, ensuring proportional representation of both classes across all splits.

Implementation details. All models were implemented in TensorFlow. Images were resized to 224×224 pixels for the first stage classification pipeline, and upsampled to 2240×2240 for the patch wise localization stage to preserve fine grained spatial detail. Training was performed on an NVIDIA GPU with a batch size of 32 and the Adam optimiser. The composite loss function used for autoencoder training is defined as:

$$\mathcal{L}_s = \alpha_1 \mathcal{L}_f + \alpha_2 \mathcal{L}_2$$

where $\mathcal{L}_f$ is a perceptual feature loss computed in VGG feature space and $\mathcal{L}_2$ is the pixel wise mean squared error. Both loss weights (α_1 , α_2) were set to 1.0 following a grid search over {0.1, 0.5, 1.0, 2.0}.

3.2 Autoencoder Architecture Comparison

To identify the optimal reconstruction backbone for anomaly detection, six architectures were evaluated under identical training conditions: a standard convolutional autoencoder (AE), a Masked Autoencoder (MAE), a Vision Transformer Autoencoder (ViT-AE), a Variational Autoencoder (VAE), and the GAN-based reconstruction models GANomaly, Skip-GANomaly, and Dense-GANomaly. The GAN based variants were included as baselines given their established use in X-ray anomaly detection literature.

Reconstruction quality was assessed using two complementary metrics: Peak Signal to Noise Ratio (PSNR), which measures pixel level fidelity, and Structural Similarity Index (SSIM), which captures perceptual similarity in terms of luminance, contrast, and structure. Anomaly discriminability was separately quantified using Mean Absolute Error (MAE) and Mean Squared Error (MSE) of the reconstruction, computed over the held out test set containing both normal and anomalous images. A well suited backbone should produce low reconstruction error on normal images (indicating faithful reconstruction of learned patterns) and high error on anomalous images (indicating failure to reconstruct unseen threat structures).

All architectures were trained for 100 epochs with early stopping based on validation loss, with the best checkpoint selected for evaluation.

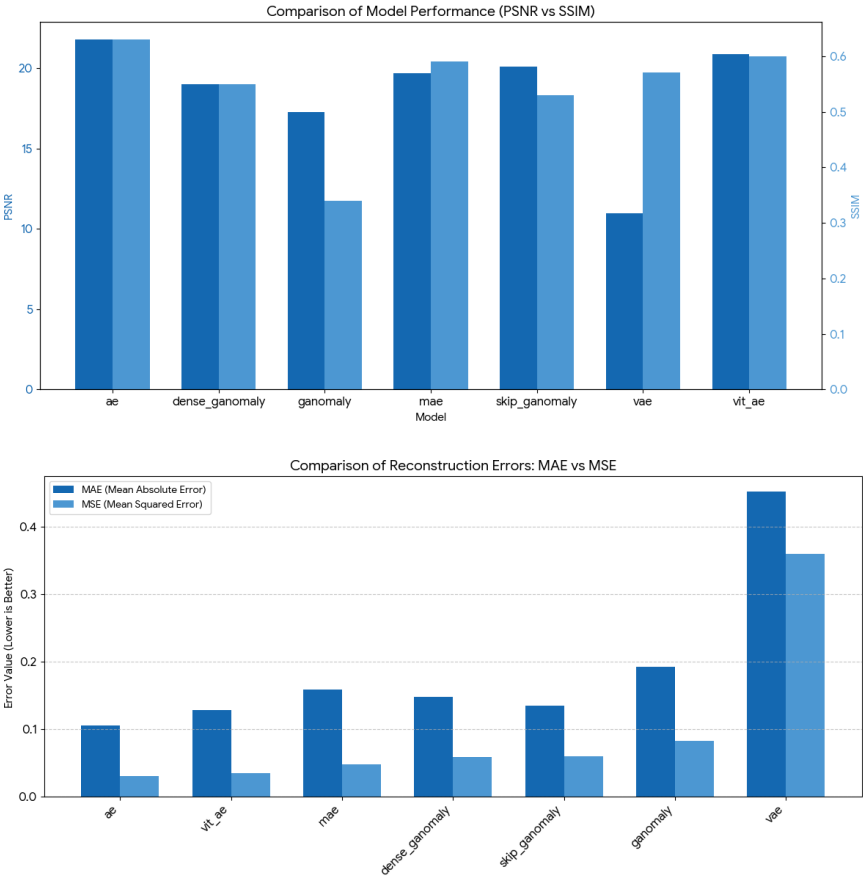


Figure 3: Encoder-Decoder performance comparison across different models

3.3 Classification Model Selection

Disparity maps produced by the selected autoencoder are, in themselves, spatial residual images rather than classification ready signals. To convert these into image level anomaly predictions, a feature extraction step was applied: statistical descriptors including the mean, standard deviation, maximum value, and histogram based features were extracted from each disparity map, forming a compact fixed length feature vector per image.

Three classical machine learning classifiers were trained and evaluated on these feature vectors: Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Logistic Regression (LR). Classical classifiers were preferred over end to end deep classifiers here because the feature vectors are low dimensional (removing the need for deep capacity), the training set is small and imbalanced (where tree based ensembles and linear models are more robust than overparameterised networks), and the resulting pipeline remains computationally lightweight and interpretable.

Each classifier was trained on the 70% training split, tuned on the 20% validation split, and evaluated on the held out 10% test split. Performance was measured using Accuracy, F1

Model	Accuracy	F1	ROC-AUC
Random Forest	0.82 – 0.99	0.42 – 0.62	0.85 – 0.92
Extreme Gradient Boosting	0.74 – 0.84	0.32 – 0.44	0.76 – 0.79
Logistic Regression	0.81 – 0.87	0.30 – 0.40	0.90 – 0.91

Table 1: Classification performance comparison across different models.

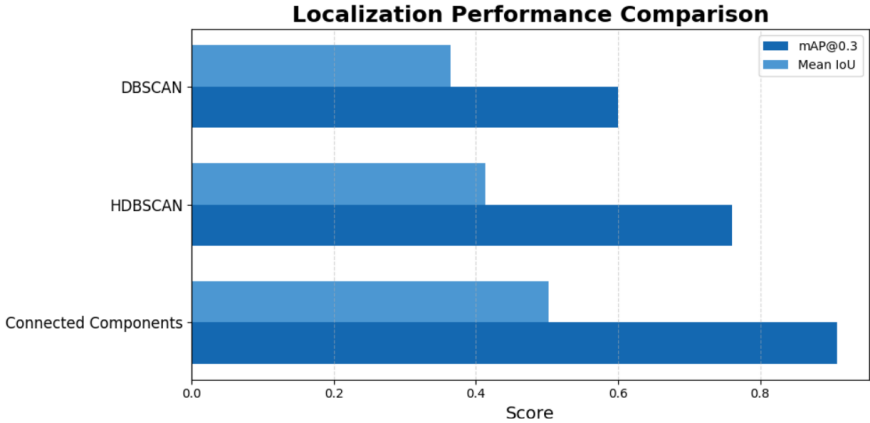


Figure 4: Localization performance comparison across different models

Score, and ROC-AUC. F1 was included alongside accuracy given the 10:1 class imbalance, where accuracy alone can be misleading. ROC-AUC was used to assess ranking ability independent of any particular classification threshold.

3.4 Localization Strategy Comparison

Once an image is classified as anomalous, the localization stage is triggered. Three spatial anomaly localization strategies were compared to identify the most effective approach for generating precise bounding box predictions:

Connected Component Analysis (CCA). The disparity map is thresholded to produce a binary mask, and connected components are extracted. Components are filtered by size and used directly to form padded bounding boxes.

DBSCAN Clustering. The coordinates of high disparity pixels are treated as a point cloud and clustered using DBSCAN, with bounding boxes formed around each identified cluster.

HDBSCAN Clustering. A hierarchical variant of DBSCAN that automatically determines the number of clusters and is more robust to varying density, applied to the same high disparity pixel coordinates.

All three strategies were applied identically on the same disparity maps produced by the selected autoencoder. Localization performance was evaluated using mean Average Precision at IoU threshold 0.3 (mAP@0.3) and Mean IoU, following conventions established in weakly supervised object detection literature and consistent with prior work on the SIXray dataset.

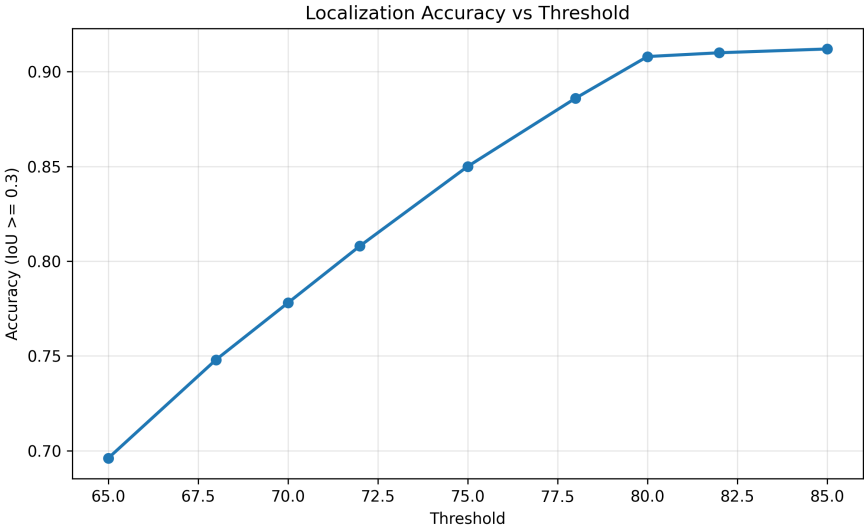


Figure 5: Threshold values experimentation

3.5 Hyperparameter Optimization for Connected Component Analysis

Given that Connected Component Analysis was identified as the superior localization strategy, its three governing hyperparameters were systematically optimised: the binarization threshold T , the minimum component size, and the maximum component area ratio.

Binarization threshold T . The threshold determines which pixels of the disparity map are considered anomalous. A sweep was conducted over $T \in \{65, 68, 70, 72, 75, 78, 80, 82, 85\}$, with localization accuracy ($\text{IoU} \geq 0.3$) recorded at each value. Thresholds that are too low admit large amounts of background noise into the binary mask, producing spurious components; thresholds that are too high suppress genuine anomaly signal in regions of moderate disparity.

Minimum component size. A minimum pixel count filter removes small connected components that are likely noise rather than true threat regions. The minimum size was swept over the range $\{8, 10, 12, 15, 18, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120\}$ pixels.

Maximum area ratio. An upper bound on the fraction of the image area that any single component may occupy prevents the entire disparity map from being flagged as a single diffuse anomaly region, a common failure mode when the reconstruction error is distributed globally rather than locally. This ratio was swept over $\{0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70\}$.

Each hyperparameter was optimised independently, holding the others at their previously determined optimal values, using a sequential grid search. Final values were selected based on peak localization accuracy on the validation set.

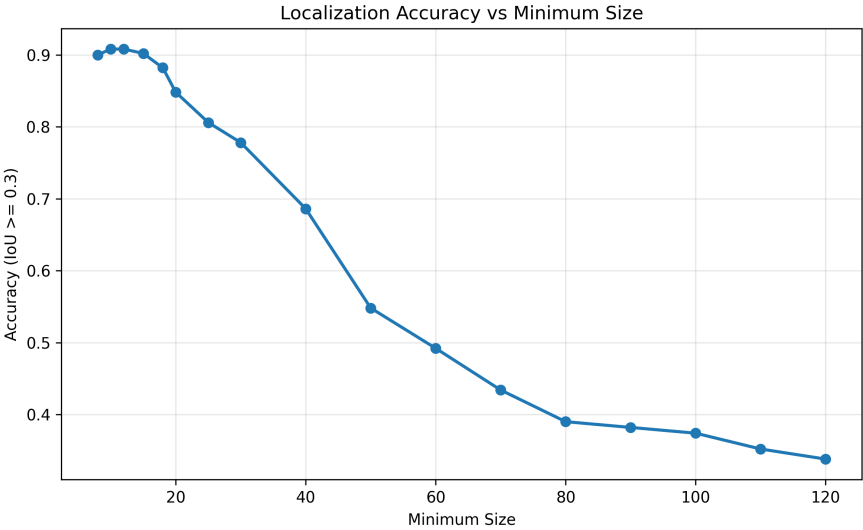


Figure 6: Minimum pixel count filter experimentation

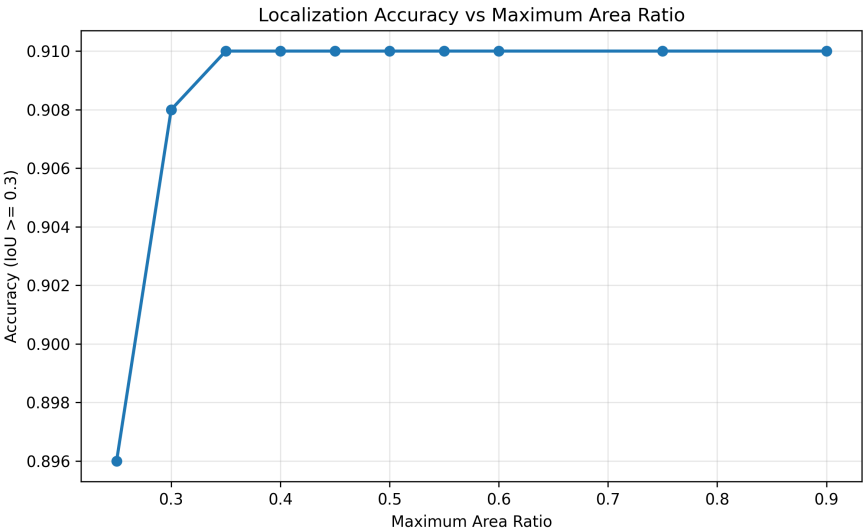


Figure 7: Maximum area ratio experimentation

4 Conclusion

This paper presented a Two-Stage Reconstruction-Guided Hybrid Framework for unsupervised anomaly detection and localization in baggage X-ray security screening. By training a Convolutional Autoencoder exclusively on normal baggage images, the proposed system avoids the annotation bottleneck that limits supervised approaches, while retaining discriminative power through a Random Forest classifier trained on features derived from the pixel-wise disparity map. The patch-wise reconstruction strategy in the second stage, combined with connected component analysis, provides fine-grained spatial localization of anomalous regions without requiring any ground-truth defect annotations at training time.

Experiments on the SIXray benchmark demonstrate that the proposed framework achieves a Detection ROC-AUC of 0.916 and a detection accuracy of 99.4%, outperforming established adversarial baselines including GANomaly, Skip-GANomaly, and Dense-GANomaly. More significantly, the framework extends beyond image-level detection to provide actionable spatial output, achieving a localization accuracy of 0.908 at $\text{IoU} \geq 0.30$ and a mean IoU of 0.502 - a capability that prior reconstruction-based methods do not report, and one that is essential for practical deployment in operator-assisted screening workflows.

These results support two broader conclusions. First, the combination of unsupervised representation learning with lightweight supervised classification represents a pragmatically effective design point: it sidesteps the data collection challenges of fully supervised systems while avoiding the instability and computational overhead of adversarial training. Second, simple, interpretable localization pipelines based on thresholded disparity masks and connected component analysis can achieve competitive spatial precision without the complexity of attention-based or segmentation-based alternatives.

Limitations and Future Work. The current framework operates on 2D single-view X-ray images, and reconstruction quality degrades in regions of heavy occlusion where multiple dense objects overlap. Future work will explore multi-view fusion and 3D Computed Tomography (CT) data to resolve ambiguities that are fundamentally irrecoverable from a single projection. Additionally, the fixed threshold applied during disparity map binarization is sensitive to image-level brightness variation; adaptive thresholding strategies or learned threshold predictors trained on reconstruction confidence will be investigated to improve robustness. Finally, integrating Continual Learning mechanisms to update the autoencoder's representation of "normal" content dynamically, without full retraining, will be essential for adapting to evolving travel trends and seasonal luggage variation in live deployment settings.

5 Template

5.1 Paper length

Papers must not exceed 14 pages (excluding references). References and any appendices are unlimited. **Only** the acknowledgement and bibliography should be *excluded* from the page count: Papers must not have altered margins and formatting from those laid down by this style guide.

The bibliography should begin immediately after the paper text. It may be of any length, within reason. It must *not* include annotations, figures, or any other paraphernalia intended to subvert the paper length requirement.

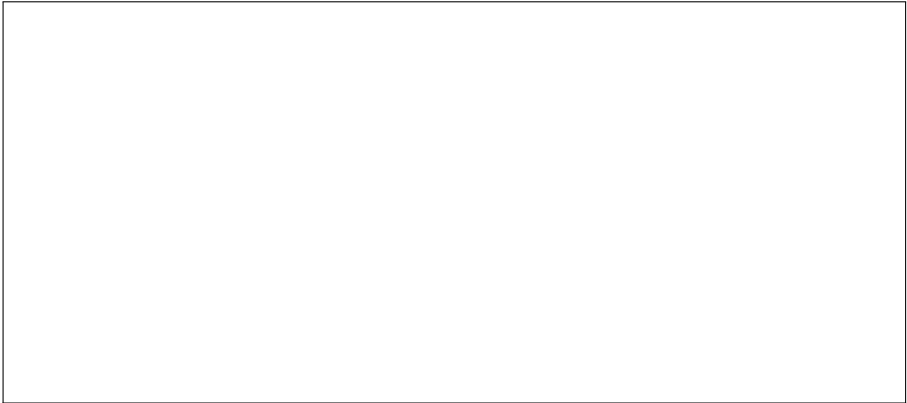


Figure 8: Example of a short caption, which should be centered.

Method	Frobnability
Theirs	Frumpy
Yours	Frobbly
Ours	Makes one’s heart Frob

Table 2: Results. Ours is better.

5.2 Citations

When citing a multi-author paper, you may save space by using “*et alia*”, shortened to “*et al.*” (not “*et. al.*” as “*et*” is a complete word.) The provided `\etal` macro is a useful *aide memoire* in this regard. However, use it only when there are three or more authors. Thus, the following is correct: “Frobnication has been trendy lately. It was introduced by Alpher [1], and subsequently developed by Alpher and Fotheringham-Smythe [2], and Alpher *et al.* [3].”

This is incorrect: “... subsequently developed by Alpher *et al.* [4] ...” because reference [4] has just two authors. If you use the `\etal` macro, then you need not worry about double periods when used at the end of a sentence as in Alpher *et al.*

We use `natbib`, so citations in random order are nicely sorted: [3, 2, 1, 4]. However, we don’t use the `compress` option, as we want each reference to have its own hyperlink and popup window.

5.3 Footnotes

Please use footnotes¹ sparingly. Indeed, try to avoid footnotes altogether and include necessary peripheral observations in the text (within parentheses, if you prefer, as in this sentence). If you wish to use a footnote, place it at the bottom of the column on the page on which it is referenced. Use Times 8-point type, single-spaced.

¹This is what a footnote looks like. It often distracts the reader from the main flow of the argument.

5.4 Mathematics

Please number all of your sections and displayed equations. It is important for readers to be able to refer to any particular equation. Just because you didn't refer to it in the text doesn't mean some future reader might not need to refer to it. It is cumbersome to have to use circumlocutions like "the equation second from the top of page 3 column 1". (Note that the ruler will not be present in the final copy, so is not an alternative to equation numbers). All authors will benefit from reading Mermin's description [10] of how to write mathematics.

5.5 References

List and number all bibliographical references in 9-point Times, single-spaced, at the end of your paper. When referenced in the text, enclose the citation number in square brackets, for example [5]. Where appropriate, include the name(s) of editors of referenced books.

5.6 Color

Color is valuable, and will be visible to readers of the electronic copy. However ensure that, when printed on a monochrome printer, no important information is lost by the conversion to grayscale.

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